

Q.1) Choose the correct answer and label them in your answer sheet and write model label (A or B).

1. The result of a SQL SELECT statement is
a. form ☐ b. table ☒ c. file ☐ d. all of these ☐
2. A _____ contains data that is derived from the database.
a. field ☐ b. view ☒ c. record ☐ d. table ☐
3. In Hypertext database, any object, whether it is a text, a picture, or a film, can be _____ to any other object.
a. linked ☒ b. moved ☐ c. integrated ☐ d. none of these ☐
4. Database improved data access to users through use of host and _____ languages.
a. high level programming ☐ b. low level programming ☐ c. Query ☒ d. None of these ☐
5. SQL includes _____
a. Database and questions ☐ b. Data manipulation language ☒ c. Both a & b ☐ d. None of these ☐
6. SELECT * FROM student WHERE name LIKE _____ is used to select all data from student table starting the name form letter 't'
a. 't%'; ☐ b. "%t%"; ☐ c. "%t"; ☒ d. "t%"; ☐
7. _____ the database is the process of storing the data itself on storage medium that is controlled by the DBMS.
a. Constructing ☒ b. Integrating ☐ c. Sharing ☐ d. Manipulating ☐
8. Advantages of DBMS in file systems include _____
a. data sharing ☐ b. No DBMS in file systems ☒ c. data abstraction ☐ d. All of these ☐
9. Related fields in a database are grouped to form a _____
a. data file ☐ b. data record ☒ c. Menu ☐ d. all of these ☐
10. Database systems are _____ to design.
a. difficult ☐ b. time-consuming ☐ c. all of these ☒ d. None of these ☐
11. IN _____, all attribute values are atomic, no repeating group and no composite attributes.
a. 1NF ☐ b. 2NF ☐ c. 3NF ☒ d. UNF ☐
12. Each column in a table represents _____ of an entity.
a. a Byte ☐ b. a Logical Element ☐ c. a physical Element ☒ d. an Attribute ☐
13. Course(course_id, sec_id, semester). Here the course_id, sec_id and semester are _____ and course is a _____.
a. Relations, Attribute ☐ b. Attributes, Relation ☒ c. Tuple, Relation ☐ d. Tuple, Attributes ☐
14. A database is a logical _____ of data with some inherent meaning.
a. SQL commands ☐ b. structures ☒ c. collection ☐ d. catalogs ☐
15. The following are components of a database except _____.
a. User data ☐ b. Meta data ☐ c. Reports ☒ d. indexes ☐
16. How many primary key can a table in database have?
a. Only one ☒ b. At least one ☐ c. More than one ☐ d. Any number of ☐
17. Which of the following operation is used if we are interested in only certain columns of a table?
a. Projection ☒ b. Selection ☐ c. Union ☐ d. Join ☐
18. Which of the following is NOT a basic element of all versions of the E-R model?
a. Entities ☐ b. Attributes ☐ c. Relationships ☒ d. Primary keys ☐
19. The intersection operator is used to get the _____ tuples.
a. Different ☐ b. Common ☒ c. All ☐ d. Repeating ☐
20. _____ combines logic programming with a relational database
a. Cloud database ☐ b. Deductive database ☒ c. Active database ☐ d. OLAP ☐
21. _____ is a stationary database that can be connected to by smart phones or PDAs
a. Hypermedia database ☐ b. Distributed database ☐ c. Mobile Database ☒ d. None of these ☐
22. _____ is a technique for improving database performance
a. Database indexing ☒ b. DDL ☐ c. DML ☐ d. DCL ☐
23. _____ joins are SQL server default.
a. Outer ☐ b. Inner ☒ c. Equi ☐ d. None of these ☐
24. Database _____ is responsible for choosing appropriate structures to represent and store this data.
a. analyst ☐ b. designer ☒ c. administrator ☐ d. None of these ☐
25. _____ are primarily static, read-only databases which store archived, historical data used for analysis.
a. Temporal database ☐ b. Knowledge base ☐ c. Spatial database ☐ d. Analytic database ☒
26. In the relational model, relationships between relations or tables are created by using
a. Composite keys ☐ b. Determinants ☐ c. Candidate keys ☐ d. Foreign keys ☒
27. _____ improve processing by using multiple central processing units (CPUs)
a. Operational database ☐ b. Distributed database ☐ c. Cloud Database ☐ d. Parallel databases ☒
28. What of the following is true about entity's attributes?
a. It is simple ☐ b. It can be composite ☐ c. all of these ☒ d. None of these ☐
29. Stand-alone end users include engineers, scientists, business analysts who thoroughly familiarize themselves with the facilities of DBMS
a. True ☒ b. False ☐

30. What distinguishes the _____ is that the database and DBMS can reside remotely and it can be maintained and developed through Web browsers and Open APIs
 a. Spatial b. Real Time c. Cloud database d. None of these
31. _____ level Unifies the various external views into global views, describe structure and constraints
 a. physical b. external c. conceptual d. None of these
32. When all of the entity type attributes are directly dependent on its primary key, it's in
 a. 1NF b. 2NF c. 3NF d. None of these
33. Graph database is considered a kind of _____
 a. Real Time databases b. DBMS c. SQL d. NoSQL
34. Functional Analysis means establishing the _____ of data.
 a. use b. nature c. complication d. None of these
35. An entity is any object in the system that we want to _____ and store information.
 a. use b. retrieve c. model d. None of these
36. The self-describing nature of a database system is stored in _____
 a. system file b. the application program c. data structure d. DBMS catalog
37. The database environment has all of the following components except:
 a. users b. separate files c. database administrator d. database administrator
38. Relational Algebra is _____
 a. Data Definition Language b. Meta Language c. Procedural query Language d. None of these
39. _____ produces the relation that has attributes of R1 and R2
 a. Cartesian product b. difference c. Intersection d. None of these
40. A functional dependency is a relationship between or among:
 a. tables b. rows c. relations d. attributes
41. Which one of the following cannot be taken as a primary key?
 a. ID b. Register number c. Dept. ID d. Street
42. The numbers of tuples in a relation is called
 a. Counter b. Order c. Degree d. Cardinality
43. In database systems, there is insulation between application programs and
 a. data b. information c. storage media d. none of these
44. Logical operators such as (AND, OR) can be used in selection operation.
 a. true b. False
45. Manipulating a database includes such functions as querying the database to retrieve specific data, updating the database to reflect changes in the _____, and generating reports from data.
 a. databases b. reports c. mini world d. None of these
46. The _____ operation, allows to find tuples that are in one relation but are not in another.
 a. union b. Intersection c. set-difference d. projection
47. Entity relationship modeling
 a. is a design tool b. Is a graphical representation of the database system
 c. Provides high-level conceptual data model d. All of these
48. The relationship between students and courses entities is
 a. 1:1 b. 1:M c. M:N d. All of these
49. Relational algebra work on one or more relations to define a new relation without _____ the original relations.
 a. storing b. changing c. subtracting d. repeating
50. _____ can include many segments shared by multiple sites.
 a. Operational database b. Spatial database c. Temporal database d. Distributed database
51. Which of the following is the correct order of keywords for SQL SELECT statements?
 a. SELECT, FROM, WHERE b. FROM, WHERE, SELECT c. WHERE, FROM, SELECT d. SELECT, WHERE, FROM
52. Which SQL function is used to count the number of rows in a SQL query?
 a. SUM() b. NUMBER() c. COUNT() d. COUNT(*)
53. Which of the following are the five built-in functions provided by SQL?
 a. COUNT, SUM, AVG, MAX, MIN b. SUM, AVG, MIN, MAX, MULT
 c. SUM, AVG, MULT, DIV, MIN d. SUM, AVG, MIN, MAX, NAME
54. If you don't specify ASC or DESC after a SQL ORDER BY clause, the following is used by default
 a. ASC b. DESC c. There is no default value d. None of the mentioned
55. In a table, if we want to list only all different value, then which SQL clause is used?
 a. SQL DISTINCT b. SQL UNIQUE c. SQL BETWEEN d. SQL Exists
56. In database life cycle the evolution stage can be performed after the operation one.
 a. true b. False
57. A language that produces a relation that can be derived using relational calculus is relationally complete.
 a. DML b. SQL c. Relational Algebra d. None of the above
58. If the value of a non-key column is dependent on the value of another non-key column, it's called
 a. Transitive dependency b. Fully dependency c. Partially dependency d. None of these
59. Which of the following should be used to find the mean of the salary?
 a. Mean(salary) b. Avg(salary) c. Sum(salary)/ Count(salary) d. Sum(salary)/ Count(staff_ID)
60. _____ is a data item that allows us to uniquely identify individual occurrences or an entity type.
 a. Attribute b. Key c. Entity d. Relationship